NAME: SAKTHI DARSHINI V.

REG NO: 192512024

ECA5801-IOE AND AI DRIVEN SYSTEM

EXPERIMENT 4:

Simulation and Comparative Performance Evaluation of Zigbee, 6LoWPAN and NB-IoT Communication Protocols for IoE Applications using NS-3

PROGRAM CODE:

```
# ================================================================
# SIMULATION AND COMPARATIVE PERFORMANCE EVALUATION OF
# Zigbee, 6LoWPAN and NB-IoT COMMUNICATION PROTOCOLS
# FOR IoE APPLICATIONS
#
# Metrics:
# 1. Packet Delivery Ratio (PDR)
# 2. Throughput
# 3. End-to-End Delay
# 4. Energy Consumption
# ================================================================

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# ------------------------------------------------
# 1. Simulation Parameters
# ------------------------------------------------

np.random.seed(42)
```

NAME: SAKTHI DARSHINI V.

REG NO: 192512024

```
NUM_PACKETS = 1000
PACKET_SIZE = 1000          # bytes
SIMULATION_TIME = 100       # seconds

protocols = ["Zigbee", "6LoWPAN", "NB-IoT"]

# ------------------------------------------------
# 2. Protocol Parameters
# ------------------------------------------------
# Values represent typical relative characteristics
# used for this comparative simulation.

parameters = {

"Zigbee": {
"data_rate": 250_000,       # bits/sec
"packet_loss": 0.08,        # 8%
"base_delay": 0.015,        # seconds
"tx_energy": 0.12,          # mJ/packet
"rx_energy": 0.08           # mJ/packet
},

"6LoWPAN": {
"data_rate": 250_000,       # bits/sec
"packet_loss": 0.05,        # 5%
"base_delay": 0.025,        # seconds
"tx_energy": 0.15,          # mJ/packet
"rx_energy": 0.10           # mJ/packet
},
```

NAME: SAKTHI DARSHINI V.

REG NO: 192512024

```
"NB-IoT": {
"data_rate": 60_000,        # bits/sec
"packet_loss": 0.02,        # 2%
"base_delay": 0.080,        # seconds
"tx_energy": 0.25,          # mJ/packet
"rx_energy": 0.15           # mJ/packet
}
}

# ------------------------------------------------
# 3. Simulation Function
# ------------------------------------------------

def simulate_protocol(protocol):

p = parameters[protocol]

# Generate packet transmission times
packet_times = np.linspace(
0,
SIMULATION_TIME,
NUM_PACKETS
)

# Random packet delivery
delivered = np.random.rand(NUM_PACKETS) > p["packet_loss"]

delivered_packets = np.sum(delivered)
```

NAME: SAKTHI DARSHINI V.

REG NO: 192512024

```
lost_packets = NUM_PACKETS - delivered_packets

# ------------------------------------------------
# Delay calculation
# ------------------------------------------------

transmission_delay = (
PACKET_SIZE * 8
) / p["data_rate"]

random_delay = np.random.normal(
loc=0,
scale=p["base_delay"] * 0.15,
size=NUM_PACKETS
)

delay = p["base_delay"] + transmission_delay + random_delay

# Keep only successfully delivered packets
successful_delay = delay[delivered]

average_delay = np.mean(successful_delay)

# ------------------------------------------------
# Packet Delivery Ratio
# ------------------------------------------------

pdr = (delivered_packets / NUM_PACKETS) * 100
```

NAME: SAKTHI DARSHINI V.

REG NO: 192512024

```
# ------------------------------------------------
# Throughput
# ------------------------------------------------

total_bits_delivered = delivered_packets * PACKET_SIZE * 8

throughput = (
total_bits_delivered / SIMULATION_TIME
) / 1000       # kbps

# ------------------------------------------------
# Energy Consumption
# ------------------------------------------------

tx_energy = NUM_PACKETS * p["tx_energy"]

rx_energy = delivered_packets * p["rx_energy"]

total_energy = tx_energy + rx_energy

# ------------------------------------------------
# Return results
# ------------------------------------------------

return {
"Protocol": protocol,
"Packets Sent": NUM_PACKETS,
"Packets Delivered": delivered_packets,
```

NAME: SAKTHI DARSHINI V.

REG NO: 192512024

```
"Packets Lost": lost_packets,
"PDR (%)": pdr,
"Throughput (kbps)": throughput,
"Average Delay (ms)": average_delay * 1000,
"Energy Consumption (mJ)": total_energy
}


# ------------------------------------------------
# 4. Run Simulation
# ------------------------------------------------

results = []

for protocol in protocols:
result = simulate_protocol(protocol)
results.append(result)

results_df = pd.DataFrame(results)

# ------------------------------------------------
# 5. Display Results
# ------------------------------------------------

print("=" * 75)
print("COMPARATIVE PERFORMANCE EVALUATION")
print("=" * 75)

display(
```

NAME: SAKTHI DARSHINI V.

REG NO: 192512024

```
results_df.round(3)
)

# ------------------------------------------------
# 6. Performance Graph - PDR
# ------------------------------------------------

plt.figure(figsize=(8, 5))

plt.bar(
results_df["Protocol"],
results_df["PDR (%)"]
)

plt.xlabel("Communication Protocol")
plt.ylabel("Packet Delivery Ratio (%)")
plt.title("Packet Delivery Ratio Comparison")

plt.ylim(0, 105)
plt.grid(axis="y", linestyle="--", alpha=0.5)

plt.show()


# ------------------------------------------------
# 7. Performance Graph - Throughput
# ------------------------------------------------

plt.figure(figsize=(8, 5))
```

NAME: SAKTHI DARSHINI V.

REG NO: 192512024

```
plt.bar(
results_df["Protocol"],
results_df["Throughput (kbps)"]
)

plt.xlabel("Communication Protocol")
plt.ylabel("Throughput (kbps)")
plt.title("Throughput Comparison")

plt.grid(axis="y", linestyle="--", alpha=0.5)

plt.show()



# ------------------------------------------------
# 8. Performance Graph - Delay
# ------------------------------------------------

plt.figure(figsize=(8, 5))

plt.bar(
results_df["Protocol"],
results_df["Average Delay (ms)"]
)

plt.xlabel("Communication Protocol")
plt.ylabel("Average End-to-End Delay (ms)")
plt.title("End-to-End Delay Comparison")
```

NAME: SAKTHI DARSHINI V.

REG NO: 192512024

```
plt.grid(axis="y", linestyle="--", alpha=0.5)

plt.show()


# ------------------------------------------------
# 9. Performance Graph - Energy
# ------------------------------------------------

plt.figure(figsize=(8, 5))

plt.bar(
results_df["Protocol"],
results_df["Energy Consumption (mJ)"]
)

plt.xlabel("Communication Protocol")
plt.ylabel("Energy Consumption (mJ)")
plt.title("Energy Consumption Comparison")

plt.grid(axis="y", linestyle="--", alpha=0.5)

plt.show()


# ------------------------------------------------
# 10. Combined Performance Comparison
# ------------------------------------------------
```

NAME: SAKTHI DARSHINI V.

REG NO: 192512024

```
fig, axes = plt.subplots(2, 2, figsize=(12, 9))

# PDR
axes[0, 0].bar(
results_df["Protocol"],
results_df["PDR (%)"]
)
axes[0, 0].set_title("Packet Delivery Ratio")
axes[0, 0].set_ylabel("PDR (%)")
axes[0, 0].grid(axis="y", linestyle="--", alpha=0.5)

# Throughput
axes[0, 1].bar(
results_df["Protocol"],
results_df["Throughput (kbps)"]
)
axes[0, 1].set_title("Throughput")
axes[0, 1].set_ylabel("kbps")
axes[0, 1].grid(axis="y", linestyle="--", alpha=0.5)

# Delay
axes[1, 0].bar(
results_df["Protocol"],
results_df["Average Delay (ms)"]
)
axes[1, 0].set_title("Average Delay")
axes[1, 0].set_ylabel("Delay (ms)")
axes[1, 0].grid(axis="y", linestyle="--", alpha=0.5)
```

NAME: SAKTHI DARSHINI V.

REG NO: 192512024

```
# Energy
axes[1, 1].bar(
results_df["Protocol"],
results_df["Energy Consumption (mJ)"]
)
axes[1, 1].set_title("Energy Consumption")
axes[1, 1].set_ylabel("Energy (mJ)")
axes[1, 1].grid(axis="y", linestyle="--", alpha=0.5)

plt.suptitle(
"Comparative Performance of Zigbee, 6LoWPAN and NB-IoT",
fontsize=14
)

plt.tight_layout()
plt.show()



# ------------------------------------------------
# 11. Best Protocol for Each Metric
# ------------------------------------------------

print("\n" + "=" * 75)
print("PERFORMANCE SUMMARY")
print("=" * 75)

best_pdr = results_df.loc[
results_df["PDR (%)"].idxmax(),
```

NAME: SAKTHI DARSHINI V.

REG NO: 192512024

```
"Protocol"
]

best_throughput = results_df.loc[
results_df["Throughput (kbps)"].idxmax(),
"Protocol"
]

lowest_delay = results_df.loc[
results_df["Average Delay (ms)"].idxmin(),
"Protocol"
]

lowest_energy = results_df.loc[
results_df["Energy Consumption (mJ)"].idxmin(),
"Protocol"
]
print("Highest Packet Delivery Ratio :", best_pdr)
print("Highest Throughput           :", best_throughput)
print("Lowest Delay                 :", lowest_delay)
print("Lowest Energy Consumption    :", lowest_energy)
```

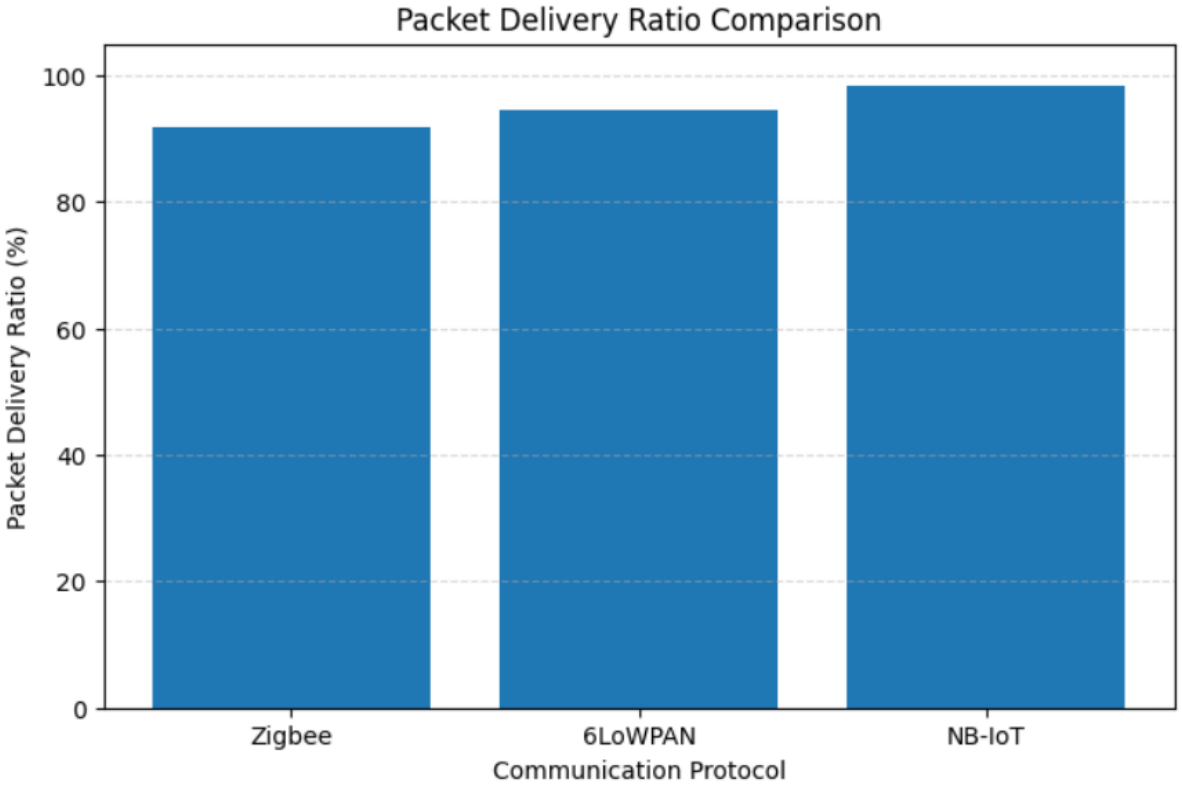
OUTPUT:

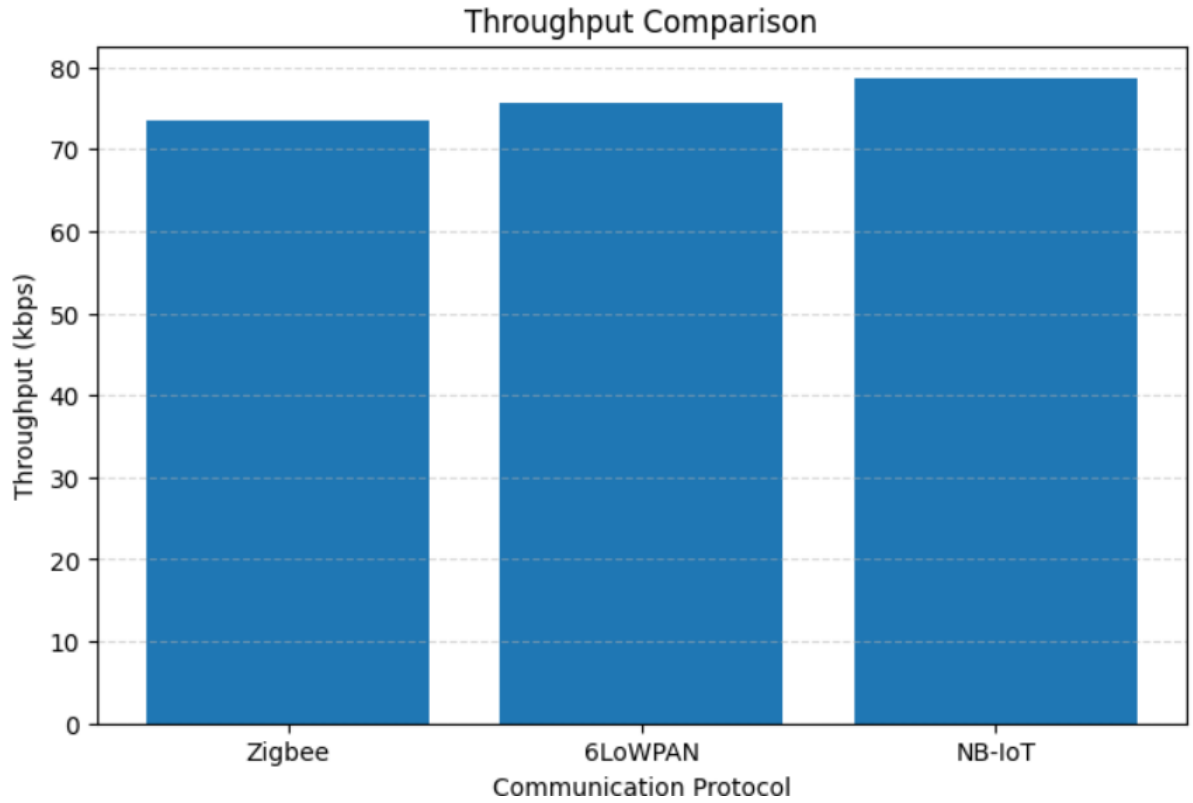
```
======================================================================
COMPARATIVE PERFORMANCE EVALUATION
======================================================================
```

	Protocol	Packets Sent	Packets Delivered	Packets Lost	PDR (%)	Throughput (kbps)	Average Delay (ms)	Energy Consumption (mJ)
0	Zigbee	1000	918	82	91.8	73.44	47.207	193.44
1	6LoWPAN	1000	946	54	94.6	75.68	56.966	244.60
2	NB-IoT	1000	983	17	98.3	78.64	213.205	397.45

NAME: SAKTHI DARSHINI V.

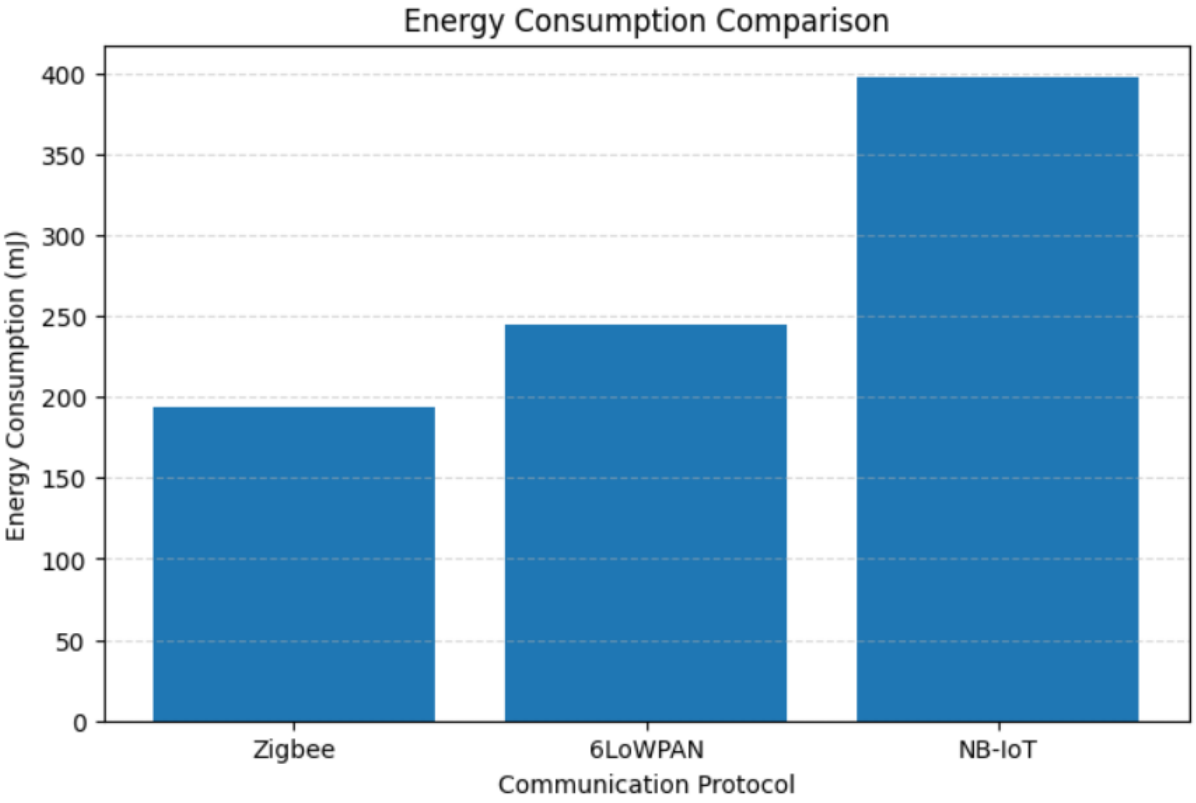
REG NO: 192512024

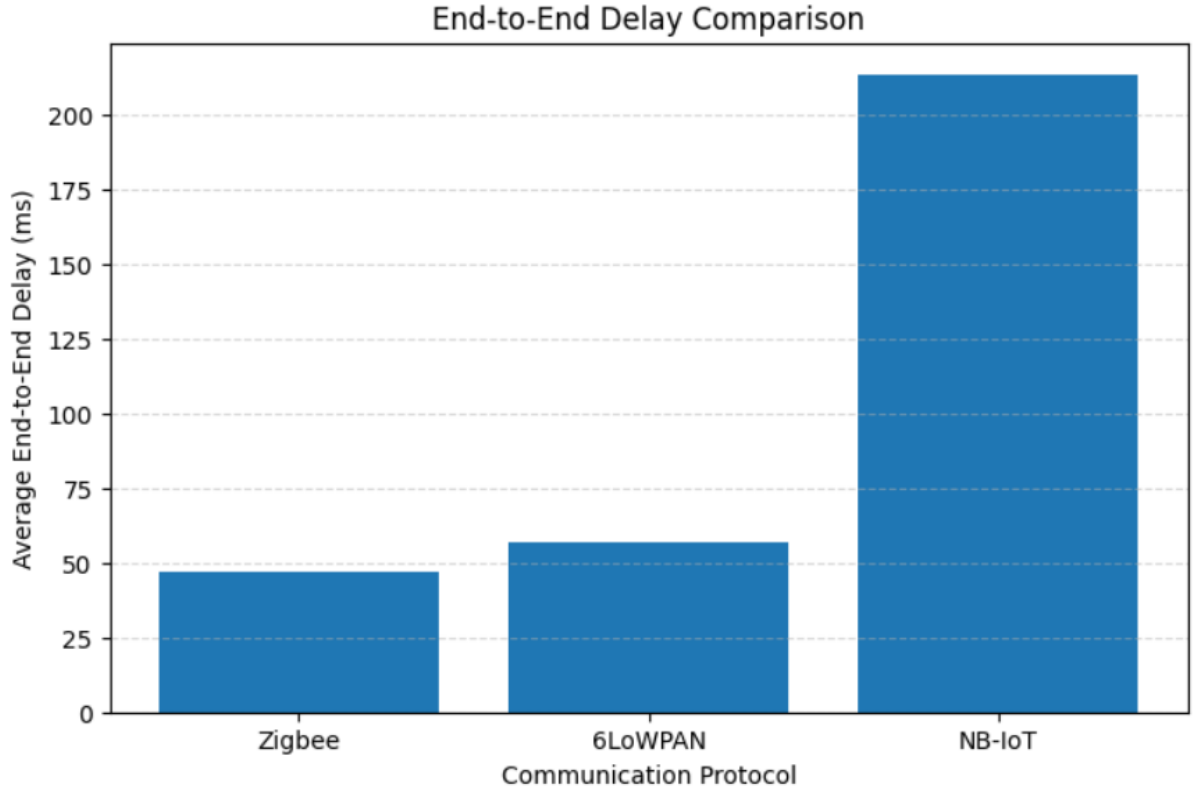




NAME: SAKTHI DARSHINI V.

REG NO: 192512024





NAME: SAKTHI DARSHINI V.

REG NO: 192512024

